WORKED SOLUTION

STEP III 2018, Question 2

Pure • Hermite Polynomials

STEP III 2018, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s3-q2

The key idea

Turn the definition round: it says that the nth derivative of z is $(-1)^n y_n z$. Differentiating this once more and comparing with the same statement for $n+1$ gives part (i) in a line. After that, part (i) is the tool that lets us differentiate a recurrence, which drives the induction in (ii). The final inequality then follows from the identity in (ii) by a second, very short induction.

Solution

Throughout, $z = \mathrm{e}^{-x^2}$, so $z' = -2xz$, and $z > 0$ for every x. Rearranging the definition,

$$\frac{\mathrm{d}^n z}{\mathrm{d}x^n} = (-1)^n y_n z \qquad (n \geq 0),$$

which also holds for $n = 0$ because $y_0 = 1$.

Part (i)

Differentiate the displayed equation, using the product rule and $z' = -2xz$:

$$\frac{\mathrm{d}^{n+1} z}{\mathrm{d}x^{n+1}} = (-1)^n (y_n' z + y_n z') = (-1)^n (y_n' - 2x y_n) z.$$

But the same equation with $n+1$ in place of n says the left-hand side is $(-1)^{n+1} y_{n+1} z$. Equating the two and dividing by $(-1)^n z$, which is never zero,

$$y_n' - 2x y_n = -y_{n+1}, \qquad \text{that is} \qquad \frac{\mathrm{d}y_n}{\mathrm{d}x} = 2x y_n - y_{n+1},$$

as required. Nothing here needed $n \geq 1$, so the result also holds for $n = 0$; we shall use that below.

For later use: $z' = -2xz$ and $z'' = (4x^2 - 2)z$, so $y_1 = 2x$ and $y_2 = 4x^2 - 2$.

Part (ii)

Base case. For $n = 1$ we need $y_2 = 2xy_1 - 2y_0$, and indeed $2x \cdot 2x - 2 = 4x^2 - 2 = y_2$.

Inductive step. Suppose that, for some $k \geq 1$,

$$y_{k+1} = 2xy_k - 2ky_{k-1}. \qquad (*)$$

We want the same relation one step further on, so we differentiate $(*)$:

$$y'_{k+1} = 2y_k + 2xy'_k - 2ky'_{k-1}.$$

? **Ask yourself** How do we get rid of the derivatives, and is part (i) allowed for y_{k-1} when $k = 1$?

Replace every derivative using part (i). For y'_{k-1} with $k = 1$ this is the $n = 0$ case, which we noted holds as well. This gives

$$2xy_{k+1} - y_{k+2} = 2y_k + 2x(2xy_k - y_{k+1}) - 2k(2xy_{k-1} - y_k),$$

so

$$y_{k+2} = 4xy_{k+1} - 4x^2y_k - 2(k+1)y_k + 4kx\,y_{k-1}.$$

The term in y_{k-1} is removed with the hypothesis $(*)$ once more: $2ky_{k-1} = 2xy_k - y_{k+1}$, so $4kx\,y_{k-1} = 4x^2y_k - 2xy_{k+1}$. Substituting,

$$\begin{aligned} y_{k+2} &= 4xy_{k+1} - 4x^2y_k - 2(k+1)y_k + 4x^2y_k - 2xy_{k+1} \\ &= 2xy_{k+1} - 2(k+1)y_k, \end{aligned}$$

which is the relation with $k+1$ in place of k. By induction, $y_{n+1} = 2xy_n - 2ny_{n-1}$ for all $n \geq 1$.

The deduction. Use the recurrence twice, once at n and once at $n+1$, to make the term $2xy_ny_{n+1}$ appear in both products:

$$\begin{aligned} y_{n+1}^2 &= y_{n+1}(2xy_n - 2ny_{n-1}) = 2xy_ny_{n+1} - 2ny_{n-1}y_{n+1}, \\ y_ny_{n+2} &= y_n(2xy_{n+1} - 2(n+1)y_n) = 2xy_ny_{n+1} - 2(n+1)y_n^2. \end{aligned}$$

Subtracting, the x terms cancel:

$$y_{n+1}^2 - y_ny_{n+2} = 2(n+1)y_n^2 - 2ny_{n-1}y_{n+1} = 2n(y_n^2 - y_{n-1}y_{n+1}) + 2y_n^2,$$

which is the required identity.

Part (iii)

Write $D_n = y_n^2 - y_{n-1}y_{n+1}$. The identity from (ii) reads

$$D_{n+1} = 2nD_n + 2y_n^2 \qquad (n \geq 1).$$

We prove $D_n > 0$ for all $n \geq 1$ by induction. For $n = 1$,

$$D_1 = y_1^2 - y_0 y_2 = 4x^2 - (4x^2 - 2) = 2 > 0$$

for every x. If $D_n > 0$ for some $n \geq 1$, then $2nD_n > 0$ because $n \geq 1$, and $2y_n^2 \geq 0$, so $D_{n+1} > 0$. Hence $y_n^2 - y_{n-1}y_{n+1} > 0$ for all $n \geq 1$ and all real x, as required.

Note that we needed D_n strictly positive to get a strict inequality: y_n^2 alone can be zero, at the zeros of y_n.

Checks

- From the recurrence, $y_3 = 2x(4x^2 - 2) - 4 \cdot 2x = 8x^3 - 12x$. Differentiating $z'' = (4x^2 - 2)z$ directly gives $z''' = (-8x^3 + 12x)z$, matching $-y_3 z$.
- Part (i) at $n = 1$: $y_1' = 2$, and $2x \cdot 2x - (4x^2 - 2) = 2$. ✓
- Part (iii) at $n = 2$: $D_2 = (4x^2 - 2)^2 - 2x(8x^3 - 12x) = 8x^2 + 4$, which is positive, and equals $2D_1 + 2y_1^2 = 4 + 8x^2$, as the identity predicts.
- A computer check of the recurrences and the inequality up to $n = 14$, at many rational values of x (including near the zeros of each y_n), agreed exactly.

Common slips

- Forgetting that the inductive step uses part (i) for y_{k-1}, which at $k = 1$ means y_0: part (i) was only asked for $n \geq 1$, so check the case $n = 0$ separately.
- Concluding only $D_{n+1} \geq 0$ in part (iii): the strict inequality comes from $D_n > 0$, not from y_n^2.

Beyond STEP

The functions y_n are the Hermite polynomials $H_n(x)$, and the relation $\mathrm{d}^n z/\mathrm{d}x^n = (-1)^n H_n z$ is their Rodrigues formula. They are orthogonal with weight e^{-x^2} on the whole real line, and they give the energy states of the quantum harmonic oscillator. The inequality in (iii) is a Turán-type inequality, a family of results that hold for many classical orthogonal polynomials. Boyce and DiPrima, *Elementary Differential Equations*, meets Hermite's equation through series solutions.

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